Cross-Cutting Exposure as an Engine of Radicalization: Platform Design, Attention, and Geography in a Stochastic Multiplex Model of Opinion Dynamics

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Do social-media platforms polarize because they hide disagreement, or because they expose people to it? We study this question in a continuous-time model in which agents diffuse through physical space by Brownian motion while interacting through an adaptive, algorithmically curated digital network, under a finite attention budget that forces digital exposure to displace physical interaction. When social influence is purely assimilative (bounded confidence), the conventional picture holds: opinion-blind long-range exposure heals locality-induced fragmentation, and algorithmic homophily builds echo chambers. But once influence includes an empirically documented but contested repulsive response to strongly opposed views, the ranking of platform designs inverts: the *neutral*, uncurated platform drives the population to maximal polarization with opinions pinned at the extremes, a controversy-seeking engagement algorithm is nearly as radicalizing, and strong algorithmic homophily—the standard culprit in the echo-chamber narrative—paradoxically shields the population from extremism. We derive this inversion analytically from a two-bloc reduction of the dynamics: the stationary fraction of attention slots pointing across blocs, set by the platform's engagement kernel, determines the radicalization rate, its arrest, and a finite-horizon crossover $\lambda_c(T)$ in the digital attention share—with no parameters fitted to the onset data—that is consistent with the simulated onset of the neutral and controversy-driven platforms and explains the absence of a sharp onset under algorithmic homophily: radicalization is rate-limited, not threshold-limited. Simulations across repulsion parameters, engagement kernels, and system sizes up to $N = 1600$ show that uncurated exposure sets the saturation level of radicalization, with curated platforms approaching that ceiling exactly as their curation stops withholding repulsive-zone content—and a controversy kernel tuned past the repulsion threshold reaching it at an even faster rate. Two further results complete the picture. First, a phase diagram in the mobility–attention plane (computed for the slowest-radicalizing, similarity-driven design) shows that once the platform owns most of an agent's attention, physical mobility becomes irrelevant to the outcome. Second, under opinion-independent mobility we detect no geographic opinion structure—a null result explained at first order by translation symmetry and tested at pair level—and a Schelling-type homophilic drift of strength χ restores geographic opinion structure only above a Péclet-number threshold $\chi\ell/D \sim 1$. The model provides a candidate mechanism for field observations in which curated cross-cutting exposure increased polarization, and implies that interventions promoting exposure diversity can have either sign depending on the prevalence of negative influence—consistent with the mixed experimental record.

I. INTRODUCTION

The most common mechanistic story about platforms and polarization runs through similarity: recommendation algorithms feed users content they already agree with, creating echo chambers whose members drift apart from the rest of society [1–3]. The story suggests an obvious remedy—expose people to the other side. Yet the best-known field experiment on this intervention found the opposite of the intended effect: Republicans exposed for a month to a stream of liberal content became substantially *more* conservative [4] (the broader field record is mixed: other interventions have reduced polarization or left it unchanged [5, 6]). Social psychology offers a candidate mechanism for such backfire—contested, as we discuss below: influence is not always assimilative, and views far beyond an agent's latitude of acceptance can repel rather than attract [7–9]. That algorithmic mediation and cross-cutting exposure can either polarize or depolarize is by now well established [10]: attraction–repulsion models show that exposure can polarize intolerant populations [11, 12]; post-distribution and recommendation algorithms can promote or mitigate polarization depending on their strategy [13–15]; and, within Physical Review E itself, mild random "nudges" toward out-group content depolarize an echo-chambered population while excessive nudging radicalizes it [16], and collaborative-filtering recommendations produce analytically tractable polarization phases [17]. What these works establish qualitatively—that the sign of an exposure intervention is not fixed—they do not reduce to a mechanism that *orders* platform designs quantitatively. That is the gap this paper addresses: we show that, under a conserved attention budget, the stationary fraction of exposure that a platform's curation places in the repulsive zone becomes the kinetic control parameter for radicalization—by which we mean, throughout, growth of the mean absolute opinion $\langle|x|\rangle$ toward the boundary of opinion space. The resulting rate theory maps any curation kernel to a state-dependent repulsive-exposure profile whose quadrature sets the radicalization timescale, and orders designs globally whenever their profiles are pointwise ordered.

This paper builds a minimal model in which that question has a sharp answer. Agents carry continuous opinions and interact through two layers simultaneously: a physical layer, where encounters are structured by geography—agents diffuse through a two-dimensional world by Brownian motion and influence each other through a short-range bounded-confidence kernel [18–20]—and a digital layer, a directed attention graph whose links are continuously rewired by a platform according to an *engagement kernel* $E(\Delta)$: the probability that content at opinion distance Δ is shown. A finite attention budget couples the layers: the digital attention share λ displaces physical interaction rather than adding to it. On top of this multiplex substrate [21] we add the two psychologically motivated ingredients: an assimilation–indifference–repulsion influence function [7, 9, 22] and heavy-tailed influence strengths.

Our headline result is an inversion. With purely assimilative influence, the model reproduces the standard picture—an opinion-blind platform acts like infinite-range mobility and heals fragmentation, while algorithmic homophily freezes fragmentation into spatially delocalized echo chambers. With the repulsive channel active, every platform radicalizes the population, but the ordering reverses: the neutral platform is the *most* polarizing, the controversy-seeking platform nearly as much, and strong algorithmic homophily the *least*, because hiding disagreement starves the repulsive channel. Uncurated exposure turns out to set the saturation level of radicalization: curated platforms approach that ceiling exactly as their curation stops withholding repulsive-zone content, and a controversy kernel tuned past the repulsion threshold reaches it at an even faster rate.

The inversion is not a simulation curiosity; it follows from the structure of the model, and we derive it in three steps. A two-bloc reduction shows that the stationary fraction of attention slots pointing across blocs, $p(y) = E(2y)/[E(0) + E(2y)]$ for blocs at $\pm y$, controls the radicalization rate $\dot{y} = 2\alpha_{\text{tot}}\lambda\eta\, p(y)\, y$; the shapes of $p(y)$ for the three kernels immediately give the ordering, the pinning of the neutral platform at the boundary, and the kinetic arrest of the controversy platform just short of it. Within this deterministic reduction, radicalization has *no* threshold in the attention share—any $\lambda > 0$ with nonzero cross-bloc exposure produces outward drift—and the reduction yields a finite-horizon crossover $\lambda_c(T)$, with no fitted parameters, whose values span two orders of magnitude across designs and match the simulated onsets of the neutral and controversy-driven platforms while predicting the absence of one for the similarity-driven platform. And integrating the fast-rewiring, well-mixed counterpart of the model reproduces the neutral $\mathrm{Var}(x)$-versus-λ curve of the spatial agent-based model nearly quantitatively, and the controversy curve for $\lambda \gtrsim 0.4$.

Two secondary results delimit the role of geography. First, in a phase diagram spanning four decades of mobility D and the full attention range λ, the radicalized region at high λ is independent of D: geography sets the social outcome only for populations whose attention it still owns. Second, under opinion-independent Brownian mobility we detect no geographic opinion structure at *any* mobility—a null result whose first-order part follows from translation invariance and that qualifies a common intuition about local echo chambers. Geographic opinion structure requires opinion–position coupling: adding a Schelling-type homophilic drift of strength χ [23, 24], we find that spatial opinion domains emerge only above a Péclet-number threshold $\chi\ell/D \sim 1$, where the homophilic drift beats diffusion over the interaction range ℓ.

The ingredients of the model each have a literature, recently systematised by Starnini *et al.* [10]: bounded confidence [18, 19, 25–27], mobile agents [20], coevolving networks [28], multiplex opinion formation [21], algorithmic bias and link recommendation [1, 2, 14], activity-driven echo chambers [3], stubborn agents [29], adaptive confidence [30], and coupled opinion–position dynamics [23, 24]. Closest in spirit is the post-distribution model of de Arruda *et al.* [13], in which an algorithm's distribution strategy determines whether polarization is promoted or mitigated under attractive and repulsive interactions; Bellina *et al.* [17] derive stationary polarization phases for collaborative-filtering recommendations, whereas our object is the *kinetics* of attraction–repulsion dynamics under conserved attention. The contribution here relative to that line is threefold: the conserved-attention multiplex construction, in which digital exposure displaces physical interaction; the reduction of an arbitrary curation kernel to a single kinetic control parameter (its stationary repulsive-zone exposure fraction), with explicit radicalization timescales; and the resulting quantitative ordering of platform designs, validated against a well-mixed limit with no parameters fitted to the onset data.

II. MODEL

A. State variables

Each of N agents carries a position $\mathbf{r}_i(t) \in [0, L)^2$ with periodic boundary conditions and an opinion $x_i(t) \in [-1, 1]$. Agent i may also carry an influence strength s_i and a stubbornness flag. The digital layer is a directed graph $A_{ij}(t) \in \{0, 1\}$, where $A_{ij} = 1$ means that the platform currently shows content by agent j to agent i. Each agent has a fixed number k of *attention slots*, $\sum_j A_{ij} = k$ for all i and t: attention is conserved, and new digital connections can only be acquired by dropping old ones.

B. Dynamics

Positions follow

$$\mathrm{d}\mathbf{r}_i = \chi\, \mathbf{f}_i\, \mathrm{d}t + \sqrt{2D}\, \mathrm{d}\mathbf{W}_i(t), \tag{1}$$

where D is the diffusion coefficient and $\mathbf{f}_i$ is an optional homophilic drift (Level 5 below; $\chi = 0$ elsewhere): the kernel-weighted mean of unit vectors pointing toward compatible neighbours ($|x_j - x_i| < \epsilon$) and away from incompatible ones. Opinions obey the Itô equation

$$\mathrm{d}x_i = \alpha_i \frac{\sum_j K(r_{ij})\, B_\epsilon(x_j - x_i)\,(x_j - x_i)}{\sum_j K(r_{ij})\, B_\epsilon(x_j - x_i)}\,\mathrm{d}t + \beta_i \frac{\sum_j A_{ij}\, s_j\, F(x_i, x_j)}{\sum_j A_{ij}\, s_j}\,\mathrm{d}t \;+\; \sigma_x\, \mathrm{d}B_i(t), \tag{2}$$

clipped to $[-1, 1]$ (a projected Itô dynamics), where r_{ij} is the minimum-image distance between agents, sums run over $j \neq i$, and the physical drift is set to zero for agents with no compatible neighbour inside the kernel cutoff—a common event in the sub-percolation regime studied below. The physical kernel is a truncated Gaussian,

$$K(r) = e^{-r^2/2\ell^2}\, \mathbf{1}(r < 3\ell), \tag{3}$$

and $B_\epsilon(\Delta) = \mathbf{1}(|\Delta| < \epsilon)$ implements bounded confidence. The truncation is not cosmetic: because the drift is normalised, an untruncated Gaussian would let an agent with no nearby compatible peers average with the entire population at $O(1)$ rate through the exponential tail, silently removing the mobility scale from the problem.

The influence function has the assimilation–indifference–repulsion form

$$F(x_i, x_j) = \begin{cases} x_j - x_i, & |x_j - x_i| < \epsilon_1, \\ 0, & \epsilon_1 \le |x_j - x_i| < \epsilon_2, \\ -\eta\,(x_j - x_i), & |x_j - x_i| \ge \epsilon_2, \end{cases} \tag{4}$$

with the repulsive branch ($\eta > 0$) active at Level 4; at Levels 2–3, F reduces to bounded confidence with threshold $\epsilon_1 = \epsilon$. This three-zone form descends from social-judgment-theory models [7, 22]. The empirical evidence for the repulsive branch is mixed: stochastic actor-oriented analyses of longitudinal network data support it [9], while controlled laboratory experiments find little evidence of negative shifts [31]; Level 4 should be read as exploring the consequences of this contested mechanism, not as assuming it settled.

C. Attention budget

The rates in Eq. (2) satisfy

$$\alpha_i = \alpha_{\mathrm{tot}}\,(1 - \lambda), \quad \beta_i = \alpha_{\mathrm{tot}}\,\lambda, \quad \lambda \in [0, 1], \tag{5}$$

so the digital attention share λ measures how much of a finite information-processing capacity is devoted to the platform. Increasing digital exposure *displaces* physical influence rather than adding to it, alongside the hard slot constraint $\sum_j A_{ij} = k$.

D. The platform: engagement-driven rewiring

In each interval $\mathrm{d}t$, agent i replaces one uniformly chosen attention slot with probability $\rho\,\mathrm{d}t$; the platform selects the replacement source j with probability

$$P(i \to j) \;\propto\; E(|x_i - x_j|)\; s_j, \tag{6}$$

where E is the platform's *engagement kernel*; the replacement is sampled among non-current sources, a finite-k correction of order k/N to the stationary exposure distribution used in Sec. III, vanishing in the mean-field limit. We compare three designs:

$$E_{\mathrm{sim}}(\Delta) = e^{-\gamma\Delta}, \qquad E_{\mathrm{neu}}(\Delta) = 1, \qquad E_{\mathrm{con}}(\Delta) = e^{-(\Delta-\delta)^2/2w^2}. \tag{7}$$

The similarity kernel models algorithmic homophily with strength γ [1, 2]; the neutral kernel is an opinion-blind baseline; the controversy kernel encodes a platform that maximises engagement by promoting content at a preferred ideological distance δ, close in spirit to the post-distribution algorithms of de Arruda *et al.* [13].

E. Model levels, heterogeneity, and observables

The model is built in five nested levels: (1) Brownian motion + bounded confidence; (2) static opinion-neutral digital links; (3) adaptive similarity-driven rewiring; (4) general engagement kernels + repulsion + heavy-tailed influence strengths (optional stubborn agents [29]); (5) homophilic mobility $\chi > 0$ added to the purely physical model of Level 1. Influence strengths are drawn as $s_i = 1 + z_i$ with z_i Lomax-distributed—density tail $p(s) \sim s^{-\kappa}$, i.e. survival exponent $\kappa - 1$—and normalised to unit mean per realisation. At the reference $\kappa = 2.5$ the variance of s is infinite: a deliberate "influencer" feature whose finite-N fluctuations play a visible role in Sec. IV E.

We measure: polarization $P = \mathrm{Var}(x)$; extremism $\langle |x| \rangle$; the number of opinion clusters n_c (gaps larger than 0.05 in sorted opinion space, groups of at least two agents); a categorical state (consensus / polarization / fragmentation for $n_c \le 1$ / $= 2$ / ≥ 3; states with $\mathrm{Var}(x) < 10^{-2}$ are additionally classed as consensus, an override that never fires in the reported data; note the scalar P and the two-cluster categorical state share the word "polarization"); the spatial–opinion correlation via Moran's I (with Gaussian spatial weights of scale ℓ) and via the *local agreement gap* $\Pr(|x_i - x_j| < \epsilon \mid r_{ij} < 3\ell) - \Pr(|x_i - x_j| < \epsilon)$; opinion assortativity ρ of the digital graph; modularity Q of the digital graph with respect to opinion clusters; and mean local disagreement $\langle |x_i - \bar{x}_{\partial i}| \rangle$ between an agent and the average opinion it is shown.

F. Numerical integration

Equations (1)–(2) are integrated by Euler–Maruyama. Unless stated otherwise, $N = 200$, $L = 1$, $\mathrm{d}t = 0.02$, $\alpha_{\mathrm{tot}} = 1$, $\epsilon = \epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$, $\ell = 0.02$,

$k = 10$, $\rho = 5$, $\gamma = 4$, $\delta = 0.8$, $w = 0.2$, and $\kappa = 2.5$. Opinion noise is $\sigma_x = 0.02$ whenever the digital layer is adaptive (Levels 3–4, including the phase diagram) and $\sigma_x = 0$ in the purely physical and static-layer experiments (Levels 1, 2, and 5). Horizons are $T = 60$ (Levels 1–3 and 5), $T = 80$ (Level 4, the robustness scans, and the attention sweeps), and $T = 50$ (the phase diagram). In system-size scans the box is rescaled as $L = \sqrt{N/200}$ so that density, kernel range and digital degree stay constant. With $\ell = 0.02$ the mean physical contact degree is $N\pi(3\ell)^2/L^2 \approx 2.3$, below the percolation threshold of the random geometric graph, so the instantaneous contact network is spatially fragmented and locality is meaningful. Statistics use 6–12 realisations per point (3 for the phase diagram); error bars denote the standard deviation across realisations. Halving dt twice shifts the Level-4 polarization values by amounts comparable to that spread ($\lesssim 0.05$, concentrated in the similarity platform and directed upward—i.e. against the inversion, which is therefore conservative) without affecting the platform ordering.

III. THEORY: TWO-BLOC REDUCTION AND MEAN-FIELD LIMIT

A. Late stage: cross-bloc attention controls radicalization

Consider the late-stage configuration observed at Level 4: two equal blocs at opinions $\pm y$. When attention slots renew quickly compared with the bloc drift (slot renewal rate ρ/k against the rate $2\alpha_{\rm tot}\lambda\eta p$ of Eq. (9) below—a condition that is only marginally met for the neutral kernel at our reference parameters, so the agreement found below also attests to the reduction's robustness against partial slot memory), the stationary probability that an attention slot of a $+$-bloc agent points into the $-$ bloc follows from Eq. (6):

$$p(y) \;=\; \frac{E(2y)}{E(0) + E(2y)}. \tag{8}$$

(Equation (8) is exact in the mean for homogeneous strengths $s_i = 1$, which is therefore the natural theoretical baseline; a homogeneous-strength control reproduces the inversion unchanged (Sec. IV D). With heavy-tailed strengths, slots are both sampled and weighted proportionally to s, and the realised cross-bloc exposure fluctuates around the homogeneous-strength prediction: for a realised population the slot-sampling probability is $E(2y)S_\mp/[E(0)S_\pm + E(2y)S_\mp]$, with $S_\pm$ the two blocs' realised strength totals, which reduces to Eq. (8) as their ratio approaches unity—guaranteed by the finite mean of the strength law, though convergence with N is slow for our infinite-variance reference distribution. Given the slot types, the expected s-weighted drift then equals the unweighted one by slot exchangeability, since strengths are independent of bloc membership; see Sec. III C.) Within-bloc content exerts no drift ($F = 0$ at $\Delta = 0$), and cross-bloc content is repulsive once $2y \geq \epsilon_2$, so the bloc positions obey

$$\frac{\mathrm{d}y}{\mathrm{d}t} \;=\; 2\,\alpha_{\rm tot}\,\lambda\,\eta\,p(y)\,y, \qquad 2y \geq \epsilon_2, \tag{9}$$

with no physical restoring force: once the blocs are separated beyond the confidence bound, the physical layer only pulls agents toward their own bloc's mean. Equation (9) is the central statement of this paper in its general form: any curation kernel E is mapped to a state-dependent cross-bloc exposure profile $p(y)$, whose quadrature sets the radicalization time [Eq. (10) below], and a global ordering of designs follows whenever their $p(y)$ profiles are pointwise ordered over the traversed opinion range—as they are for the three named kernels of Eq. (7), which are examples chosen to span the design space, not the scope of the result. Their $p(y)$ profiles order them:

- **Neutral:** $p \equiv 1/2$. The drift $\alpha_{\rm tot}\lambda\eta y$ never shuts off; opinions are driven to the boundary and pinned at ± 1.
- **Controversy-driven:** $p(y) \approx 1$ for $2y \gtrsim \epsilon_2$ (cross-bloc content is exactly what engages), so separation is fast; but as $2y \to 2$ both $E(2y)$ and $E(0)$ are far from the engagement peak and $p \to E(2)/[E(0) + E(2)] \approx e^{2(\delta-1)/w^2}$, which for $\delta < 1$ is exponentially small. The dynamics is *kinetically arrested* just short of the boundary—the platform stops pushing once the blocs are so far apart that neither identical nor opposed content engages.
- **Similarity-driven:** $p(y) = e^{-2\gamma y}/(1 + e^{-2\gamma y})$ is small for $\gamma y \gtrsim 1$. Radicalization proceeds, but at a rate suppressed by $e^{-2\gamma y}$: a slow outward creep fed by the few cross-cutting links that stochastic rewiring keeps producing.

These three behaviours are exactly what the simulations show (Sec. IV C): pinning at ± 1 for neutral ($\langle|x|\rangle \approx 1.00$), arrest slightly short of the boundary for controversy (≈ 0.95), and slow partial radicalization for similarity (≈ 0.79 at $T = 80$). Note also that within the deterministic reduction Eq. (9) has no positive critical λ: any $\lambda > 0$ with nonzero cross-bloc exposure produces outward drift. What varies across designs—by orders of magnitude—is the rate, which we quantify next.

B. Onset: radicalization is rate-limited, not threshold-limited

One might expect a critical attention share below which the physical layer's assimilative pull protects the population. A linear stability analysis of the *uniform* state indeed predicts such thresholds (balancing the inward pull $(1 - \lambda)\epsilon/2$ on edge agents against the

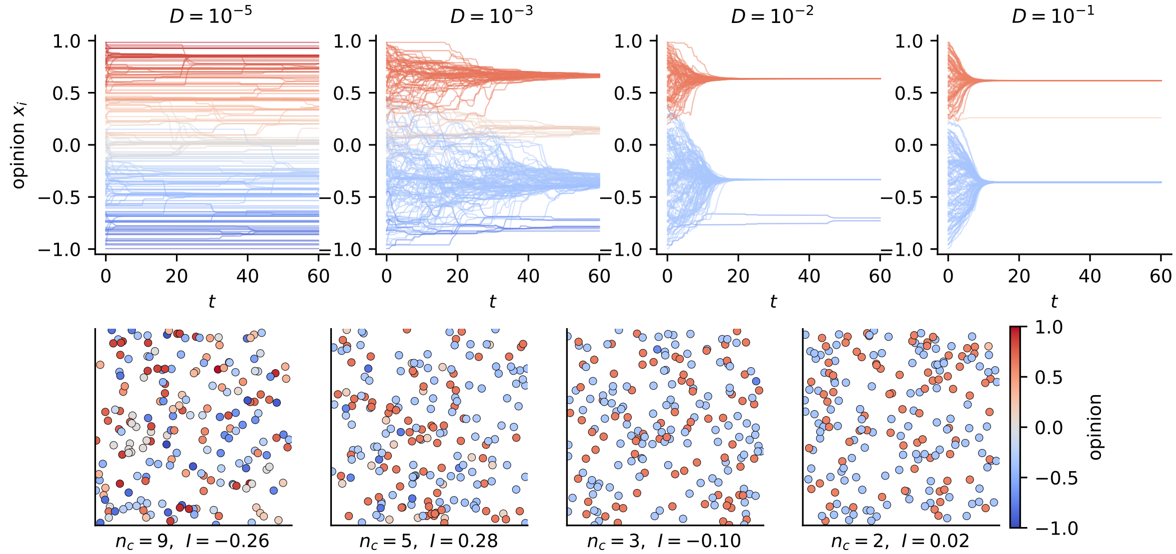

FIG. 1. Purely physical dynamics (Level 1). Top: opinion trajectories $x_i(t)$ for increasing diffusion coefficient D; colours encode final opinions. Bottom: final spatial configurations. Panel captions report the cluster count n_c and Moran's I of the single realisation shown; individual-run values of I fluctuate around a zero ensemble mean (s.d. ≈ 0.13 across seeds, single-run extremes up to $|I| \approx 0.35$; Fig. 2). Low mobility freezes many coexisting clusters; high mobility recovers the mean-field bounded-confidence outcome. Note the absence of spatial colour domains at any D.

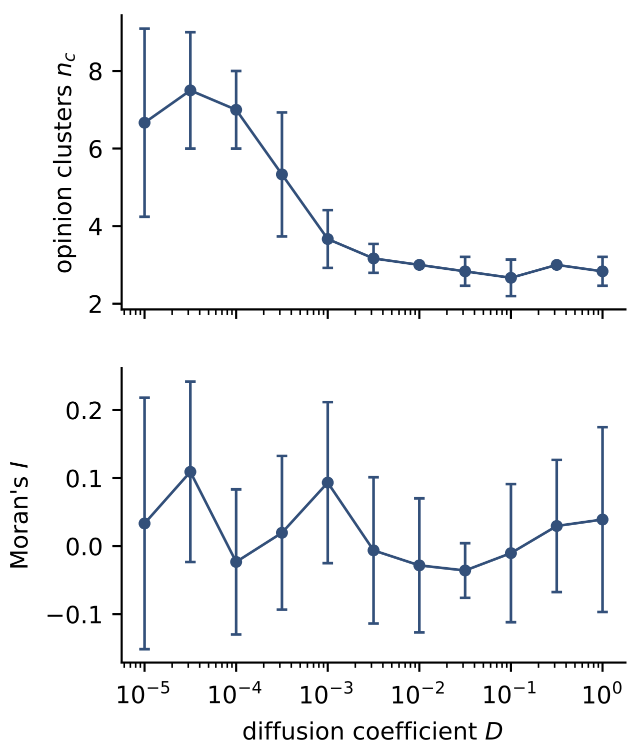

FIG. 2. Level-1 sweep over the diffusion coefficient. Top: mean number of surviving opinion clusters. Bottom: Moran's I between opinion and position (mean $\pm$ s.d. over 6 realisations). Mobility interpolates between local fragmentation and mean-field behaviour while the spatial–opinion correlation stays at noise level throughout.

engagement-weighted outward push, one obtains $\lambda^* \approx 0.34$ for the neutral and 0.55 for the controversy kernel). The simulations falsify this prediction: radicalization onsets already at $\lambda \approx 0.05$ (Sec. IV E). The reason is that the uniform state never gets to apply the analysis. Bounded-confidence fragmentation forms first, on the fast timescale $\sim (1-\lambda)^{-1}$, producing blocs near $\pm y_0$ with $y_0 \approx 0.6$ (the bounded-confidence cluster positions at $\epsilon = 0.3$, read off from Figs. 1 and 3, not fitted to the threshold data)—and since $2y_0 > \epsilon_2$, the fragmented state is unstable to repulsion for *any* $\lambda > 0$: once separated beyond the confidence bound, the physical layer exerts no restoring force between blocs (Sec. III A).

What remains is a rate. Equation (9) is separable, so the time to traverse from y_0 to y_f follows exactly by quadrature,

$$t(y_0 \to y_f) \;=\; \frac{1}{2\,\alpha_{\rm tot}\,\lambda\,\eta} \int_{y_0}^{y_f} \frac{\mathrm{d}y}{y\,p(y)}, \tag{10}$$

which maps any curation kernel to a radicalization timescale through its exposure profile alone. With the constant-exposure approximation $p(y) \approx p_0 = p(y_0)$ this gives the closed form

$$t_{\rm rad}(\lambda) \;=\; \frac{\ln(y_f/y_0)}{2\,\alpha_{\rm tot}\,\lambda\,\eta\,p_0}, \tag{11}$$

and the apparent onset in any finite observation window T is the crossover $t_{\rm rad} = T$:

$$\lambda_c(T) \;=\; \frac{\ln(y_f/y_0)}{2\,\alpha_{\rm tot}\,\eta\,p_0\,T}. \tag{12}$$

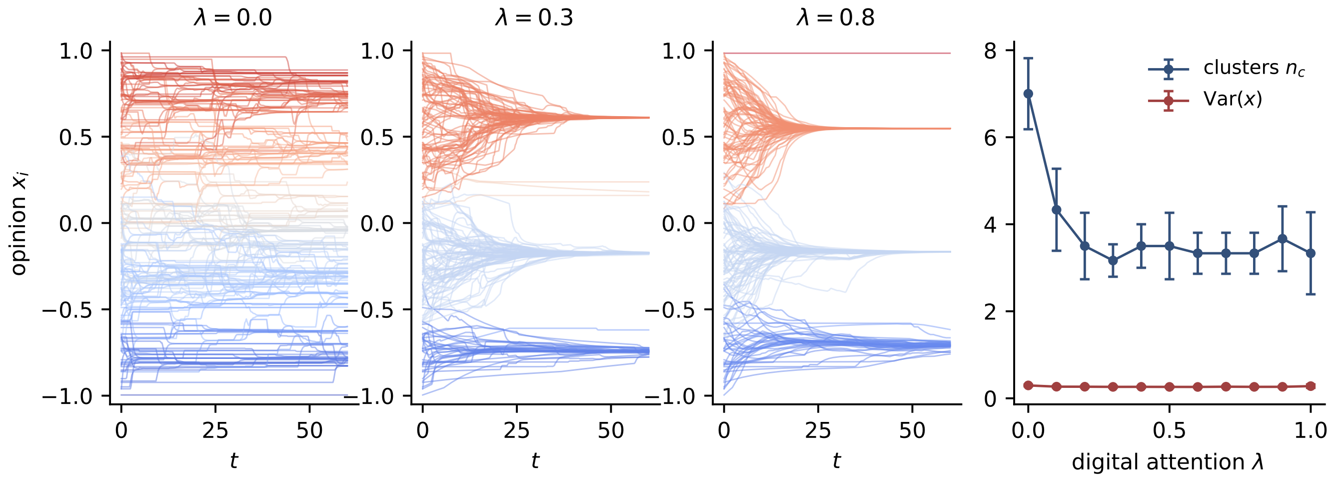

FIG. 3. Level 2: static random digital layer at low mobility ($D = 10^{-4}$). Left three panels: opinion trajectories for increasing digital attention share λ. Right: cluster count and polarization versus λ (mean ± s.d.).

Table I evaluates Eqs. (11)–(12) at the reference parameters with no fitting. The cross-bloc slot fraction p_0 spans two orders of magnitude across the designs, and with it the crossover: for the neutral and controversy platforms $\lambda_c \approx 0.01$–0.02—any noticeable digital attention radicalizes within the horizon—while for the similarity platform $\lambda_c \approx 1$: even full digital attention only partially radicalizes the population in the same window, which is exactly the smooth sub-ceiling growth observed in the simulations. "Protection" by homophilic curation is therefore kinetic, not asymptotic: it multiplies the time to reach the extremes by $p_0^{-1} \sim e^{2\gamma y_0}$, without changing the destination.

Two caveats delimit Table I. The constant-p_0 approximation is exact for the neutral kernel but conservative for the similarity kernel, whose $p(y)$ decays further as the blocs separate: evaluating the exact quadrature Eq. (10) gives radicalization times severalfold longer than $78.2/\lambda$, so the true kinetic protection is even stronger than the tabulated crossover suggests. For the controversy kernel, $p(y)$ collapses as $2y \to 2$ (Sec. III A), so t_{rad} measures the fast approach to the arrest plateau rather than to the boundary itself, which is not reached on any accessible horizon.

TABLE I. Two-bloc rate analysis, Eqs. (8) and (11)–(12), at the reference parameters ($y_0 = 0.6$, $y_f = 1$, $\eta = 0.4$, $\gamma = 4$, $\delta = 0.8$, $w = 0.2$, $T = 80$, $\alpha_{\mathrm{tot}} = 1$). No parameters are fitted to these data; y_0 is the observed bounded-confidence cluster position. See the end of Sec. III B for the caveats attached to the similarity and controversy rows.

Platform	p_0	$t_{\mathrm{rad}}(\lambda)$	$\lambda_c(T{=}80)$
Similarity-driven	0.0082	$78.2/\lambda$	0.98
Neutral	0.500	$1.28/\lambda$	0.016
Controversy-driven	0.998	$0.64/\lambda$	0.008

C. Well-mixed limit

In the fast-rewiring, well-mixed counterpart of the model, each agent feels the global bounded-confidence mean with weight $1 - \lambda$ and a population-wide engagement-weighted influence average with weight λ,

$$\mathrm{d}x = (1-\lambda)\,\frac{\mathbb{E}\big[(x'-x)\,\mathbf{1}(|x'-x|<\epsilon)\big]}{\mathbb{P}\big(|x'-x|<\epsilon\big)}\,\mathrm{d}t + \lambda\,\frac{\sum_j s_j^2\,E(|x_j-x|)\,F(x,x_j)}{\sum_j s_j^2\,E(|x_j-x|)}\,\mathrm{d}t + \sigma_x\,\mathrm{d}B, \quad (13)$$

where x' is an independent draw from the population law and the digital sum runs over the whole population. The squared strengths arise because slots are both *sampled* and *weighted* proportionally to s. The status of Eq. (13) as $N \to \infty$ depends on the tail of the strength law. For $\kappa > 3$, $\mathbb{E}[s^2]$ is finite, the self-normalised digital drift self-averages, strengths drop out entirely (they are independent of opinions), and Eq. (13) converges to a deterministic McKean–Vlasov dynamics. At our reference $\kappa = 2.5$, however, $\mathbb{E}[s^2] = \infty$ and the size-biased slot-strength law has infinite mean, so the digital drift does *not* self-average: it remains a random variable dominated by the few largest strengths at any population size—influencer fluctuations are a bulk feature of this model, not a finite-size correction. What survives for any $\kappa > 1$ is the annealed statement: by exchangeability

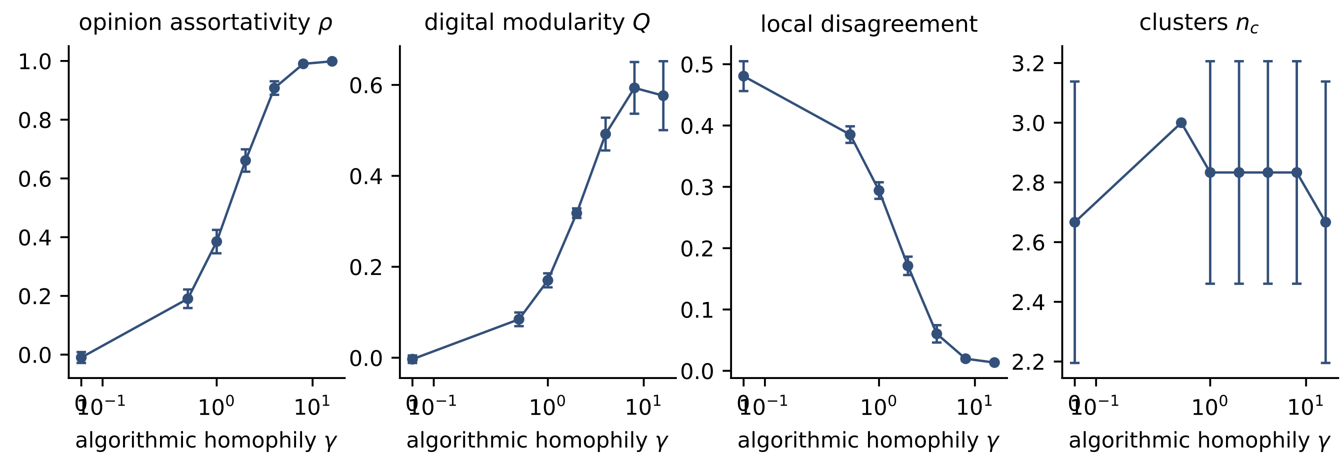

FIG. 4. Level 3: adaptive digital layer with similarity-driven recommendation, $\lambda = 0.5$, $D = 10^{-3}$. From left to right: opinion assortativity of the digital graph, modularity with respect to opinion clusters, mean local disagreement, and cluster count as functions of the algorithmic homophily γ (mean $\pm$ s.d. over 6 runs).

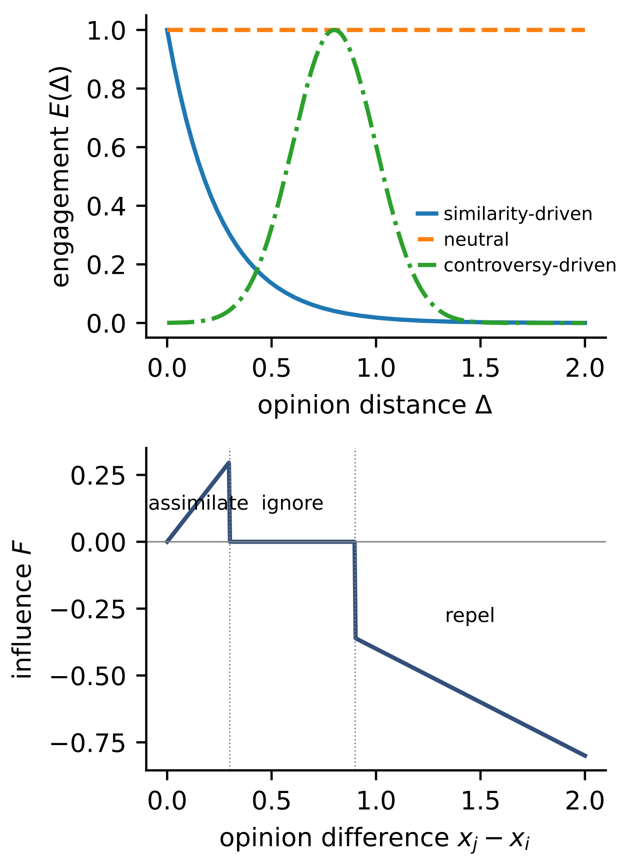

FIG. 5. Level-4 ingredients. Top: the three engagement kernels of Eq. (7) ($\gamma = 4$, $\delta = 0.8$, $w = 0.2$). Bottom: the assimilation–indifference–repulsion influence function of Eq. (4) ($\epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$).

of the slots, the *expected* drift equals the strength-free value $\mathbb{E}[EF]/\mathbb{E}[E]$ used in the two-bloc reduction, which is why Table I needs no reference to the strength law.

We therefore compare the spatial agent-based model against a direct integration of Eq. (13) as a spaceless particle system ($M = 600$, four realisations, same strength law), i.e. the model stripped of space and of the finite attention window, in Sec. IV E: with no fitted parameters it reproduces the neutral curve nearly quantitatively, the controversy-driven curve for $\lambda \gtrsim 0.4$, and tracks the shape of the similarity-driven curve, which it systematically underestimates because the slot-level sampling noise it discards (each spatial agent draws only $k = 10$ sources) is what feeds the similarity platform's slow creep. The inversion is thus a property of the opinion dynamics and the engagement weighting, not an artefact of space or slow rewiring.

IV. RESULTS

A. Baselines: mobility and neutral connectivity (Levels 1–2)

Figures 1–3 establish the assimilative baselines. In the purely physical model, the competition between the local convergence time $\tau_{\rm op} \sim \alpha_{\rm tot}^{-1}$ and the neighbourhood-renewal time $\tau_\ell \sim \ell^2/D$ controls how many opinion clusters survive: $n_c \approx 7{\pm}2$ frozen local clusters for $D \lesssim 10^{-4}$, merging into the mean-field cluster set ($n_c \approx$ 2–3 at $\epsilon = 0.3$ [18]) for $D \gtrsim 10^{-2}$, consistent with mobile-agent simulations [20]. Adding a *static, opinion-blind* digital layer at low mobility heals this excess fragmentation: a modest attention share ($\lambda \approx$ 0.2–0.3) already merges the frozen clusters down to $n_c \approx$ 3.3–3.5, close to the mean-field set, beyond which more digital attention changes nothing (Fig. 3). Opinion-blind long-range ex-

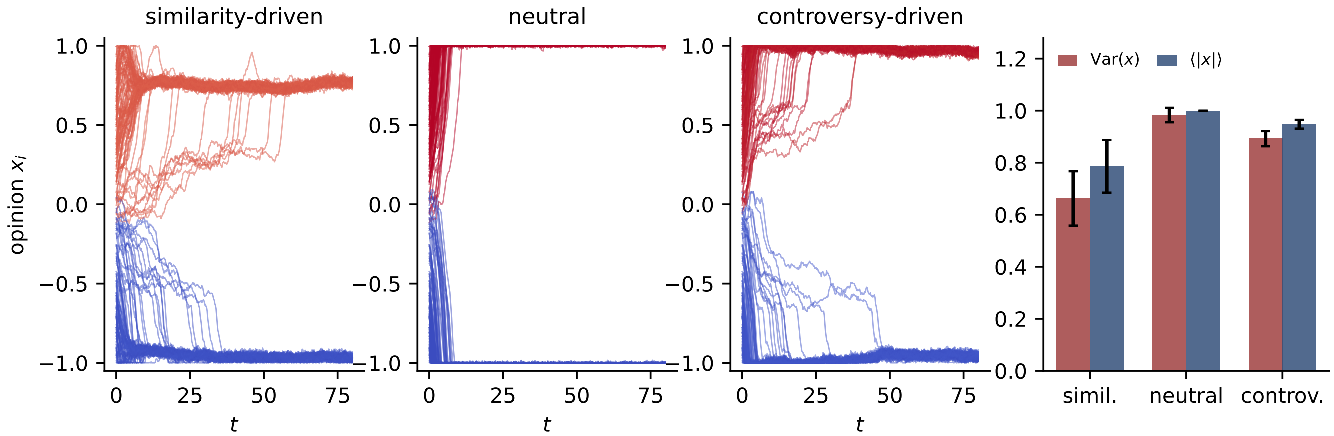

FIG. 6. Level 4: platform designs compared under repulsive influence and heavy-tailed influence strengths ($\lambda = 0.6$). Left three panels: representative opinion trajectories. Right: polarization $\mathrm{Var}(x)$ and extremism $\langle|x|\rangle$ (mean $\pm$ s.d. over 24 runs).

posure acts, in effect, like infinite-range mobility. (These levels run at $\sigma_x = 0$; repeating the Level-1 sweep at the adaptive levels' noise $\sigma_x = 0.02$ reduces the frozen cluster count—noise merges micro-clusters, $n_c = 4.8 \pm 1.6$ at $D = 10^{-5}$—but leaves the fragmentation-to-mean-field crossover intact, so the level comparison is not a noise artefact.) In the assimilative world, connectivity is benign; everything that follows is about what curation and repulsion do to this baseline.

B. Algorithmic homophily builds delocalised echo chambers (Level 3)

With similarity-driven rewiring (still purely assimilative influence), the feedback loop opinion $\to$ topology $\to$ exposure $\to$ opinion closes as γ grows (Fig. 4): opinion assortativity climbs to one, digital modularity rises, and the disagreement visible in an agent's feed collapses while global fragmentation persists. This is an echo chamber in the precise sense of Baumann *et al.* [3]—high global variance, near-zero visible disagreement—via the mechanism of Sîrbu *et al.* [1] and Santos *et al.* [2] transplanted into a coevolving multiplex [28]. The communities are spatially delocalised: geography carries no trace of them. In the assimilative world this is the worst a platform can do in our model: freeze moderate fragmentation and hide it from its participants.

C. The inversion: uncurated exposure radicalizes (Level 4)

Activating the repulsive branch and heavy-tailed influence strengths (Fig. 5) produces the inversion (Fig. 6). All three designs end with two antagonistic blocs, but with sharply different degree and speed, in the order predicted by the two-bloc reduction of Sec. III A. The neutral platform is the most polarizing ($\mathrm{Var}(x) = 0.98 \pm 0.03$ over 24 realisations, opinions pinned at ± 1 within a few time units): with $p = 1/2$, half of every feed is repulsion-triggering at all times. The controversy platform is nearly as extreme (0.89 ± 0.03) but arrests just short of the boundary ($\langle|x|\rangle \approx 0.95$), as predicted by the collapse of $p(y)$ when $2y \to 2$. The similarity platform is the least polarizing (0.66 ± 0.10): hiding opposed content starves the repulsive channel, leaving only the slow creep fed by residual cross-cutting links. Because the influence strengths have infinite variance, we also checked the distribution shape: medians (0.99/0.90/0.68) and interquartile ranges give the same ordering as the means, with no overlap between platforms. The network signatures decouple from the opinion signatures: the neutral platform reaches maximal polarization with *zero* opinion assortativity ($\rho \approx 0$)—polarization without echo chambers—whereas both curated platforms end almost perfectly assortative ($\rho \approx 0.9$–1.0).

D. Robustness: repulsive-zone exposure sets the ceiling

Figure 7 tests the ordering across the repulsion and curation parameters. Across the full (η, ϵ_2) grid the order neutral $\geq$ controversy $>$ similarity holds in the mean at every grid point—resolved by up to ≈ 7 pooled standard deviations at the widest repulsive zone ($\epsilon_2 = 0.7$), with the gaps narrowing as ϵ_2 grows until the controversy–similarity gap falls within the seed spread at $\epsilon_2 = 1.3$; all platforms polarize less as ϵ_2 grows (repulsion becomes harder to trigger) and more as η grows. The two curated platforms interpolate toward the neutral level in exactly the way the theory predicts. The similarity platform approaches the neutral bound as $\gamma \to 1$ (curation too weak to withhold repulsive content) and its protective effect saturates for $\gamma \gtrsim 8$. The controversy platform sits be-

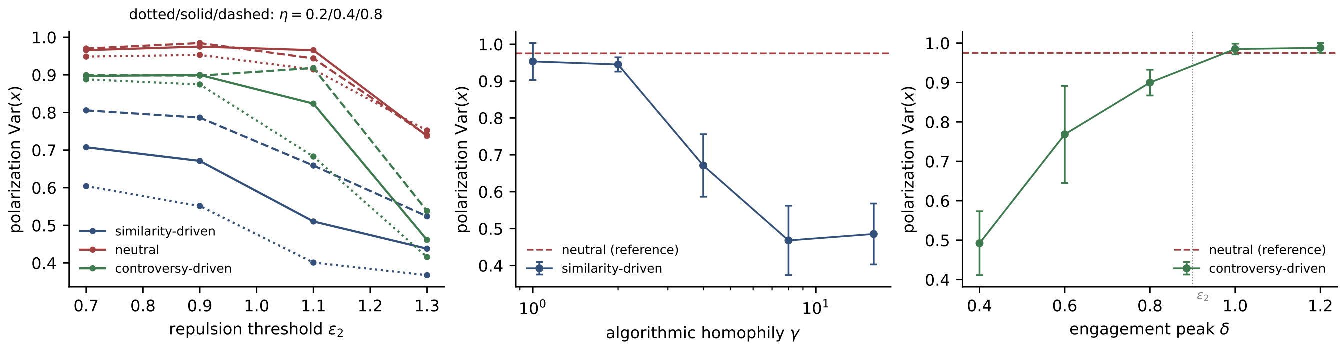

FIG. 7. Robustness of the inversion. Left: final polarization versus the repulsion threshold ϵ_2 for $\eta = 0.2/0.4/0.8$ (dotted/solid/dashed); colours denote platforms (blue similarity, red neutral, green controversy). Centre: similarity platform versus algorithmic homophily γ, with the neutral platform as reference. Right: controversy platform versus engagement peak δ; the dotted line marks $\delta = \epsilon_2$. Mean $\pm$ s.d. over 6 runs.

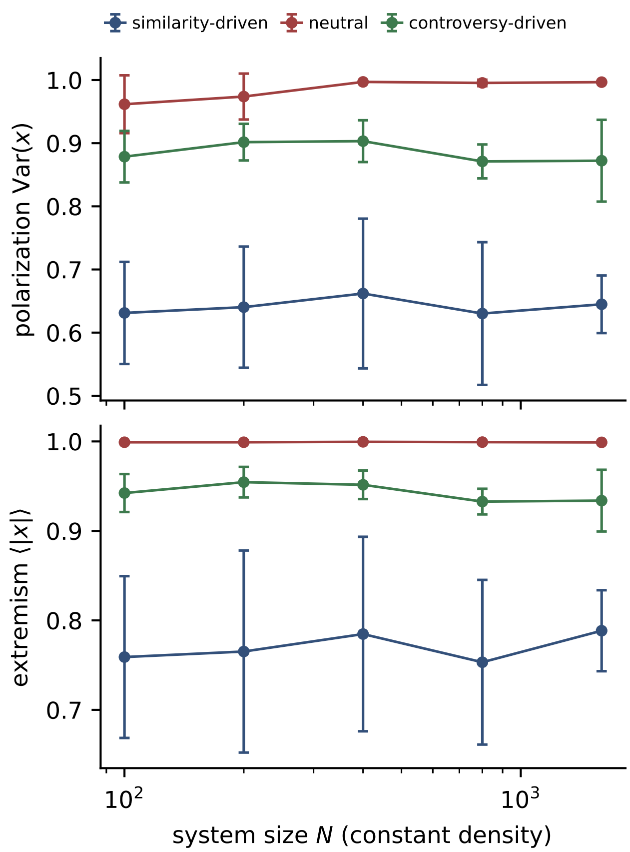

FIG. 8. System-size dependence of the inversion at constant density ($L = \sqrt{N/200}$), $\lambda = 0.6$. Polarization (top) and extremism (bottom) for the three platforms, $N = 100$–1600.

low the neutral bound while its engagement peak $\delta < \epsilon_2$ (much of its exposure lands in the indifference band) and reaches the bound once $\delta \geq \epsilon_2$, when engagement maximisation and repulsion maximisation coincide—at which point it matches the neutral end state while radicalizing at twice the neutral rate (Table I). Uncurated exposure thus sets the *saturation level* of radicalization, approached from below as curation stops withholding repulsive-zone content: a platform is exactly as dangerous as the repulsive-zone exposure it fails to withhold.

Figure 8 scans the system size at constant density up to $N = 1600$. The three platforms' polarization and extremism are essentially flat in N: the inversion is a bulk property, not a finite-size artefact.

Table II adds five further controls at the reference point ($\lambda = 0.6$, 12 realisations each). The inversion survives, with the same ordering: (i) homogeneous strengths $s_i = 1$—the regime in which the two-bloc reduction is exact in the mean—so neither the heavy tail nor influence heterogeneity is needed for the effect; (ii) a finite-variance strength law ($\kappa = 4$); (iii) reflecting instead of clipping boundaries, so the saturation ordering is not an artefact of the projection; (iv) halved and doubled attention windows ($k = 5, 20$), which shift the similarity platform's creep in the direction predicted by the slot-fluctuation mechanism (fewer slots, fewer cross-bloc bursts); and (v) slower and faster rewiring ($\rho = 1, 20$). The last is the most informative: fast rewiring strengthens homophilic protection (Var $= 0.54$ at $\rho = 20$) while slow rewiring weakens it (0.87 at $\rho = 1$, where the controversy–similarity gap narrows into the seed spread)—exactly the dependence expected from the quasi-stationarity assumption behind Eq. (8), since slowly renewed feeds let occasional cross-bloc links persist and act for longer.

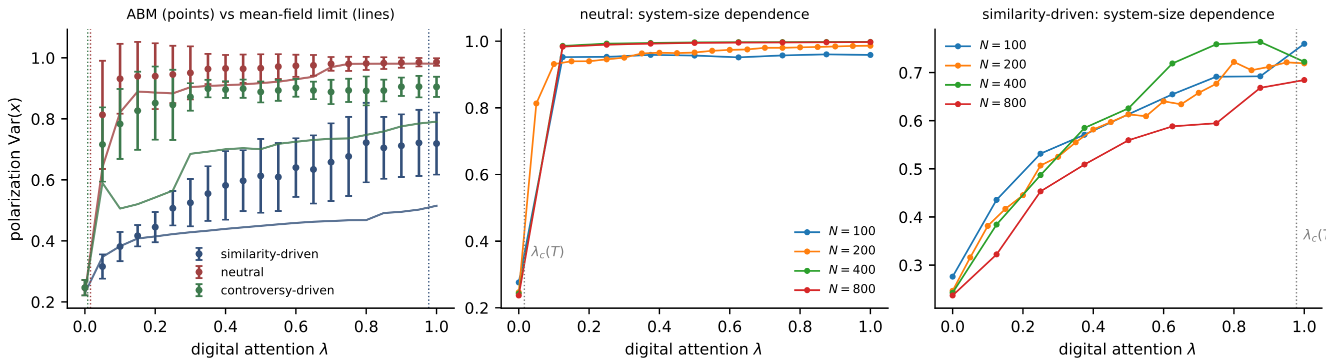

FIG. 9. Radicalization onset in the digital attention share. Left: final polarization versus λ for the three platforms; points are the spatial ABM ($N = 200$, 12 seeds), solid lines the well-mixed limit of Eq. (13), dotted verticals the finite-horizon crossovers $\lambda_c(T)$ of Table I (no parameters fitted to these data). Centre and right: system-size dependence of the Var(x)–λ curves for the neutral and similarity platforms.

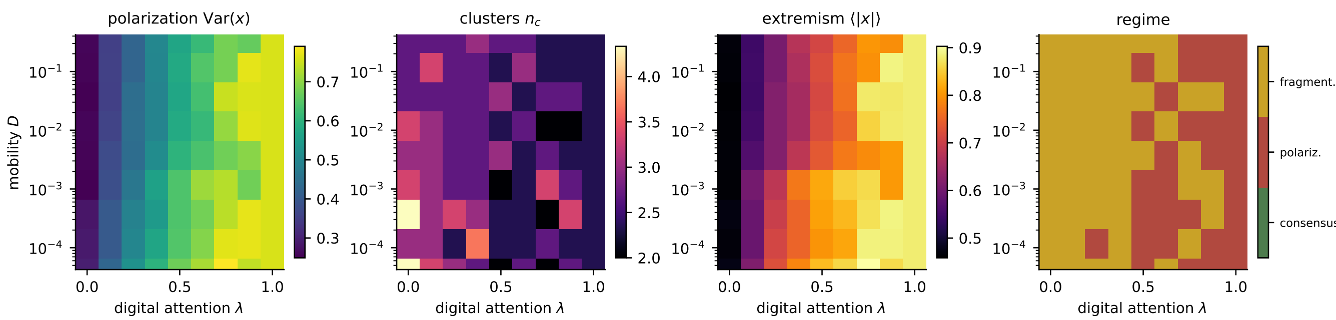

FIG. 10. Phase diagram of the full Level-4 model (similarity-driven engagement, repulsion, heavy-tailed influence) in the mobility–digital-attention plane. From left to right: polarization Var(x), cluster count n_c, extremism $\langle |x| \rangle$, and the majority categorical state over 3 realisations per grid point. The panel is a coarse regime map intended to reveal qualitative structure; at 3 realisations per point the boundaries between regimes are not sharply resolved. At $\lambda = 1$ the physical layer is off and the dynamics is independent of D by construction.

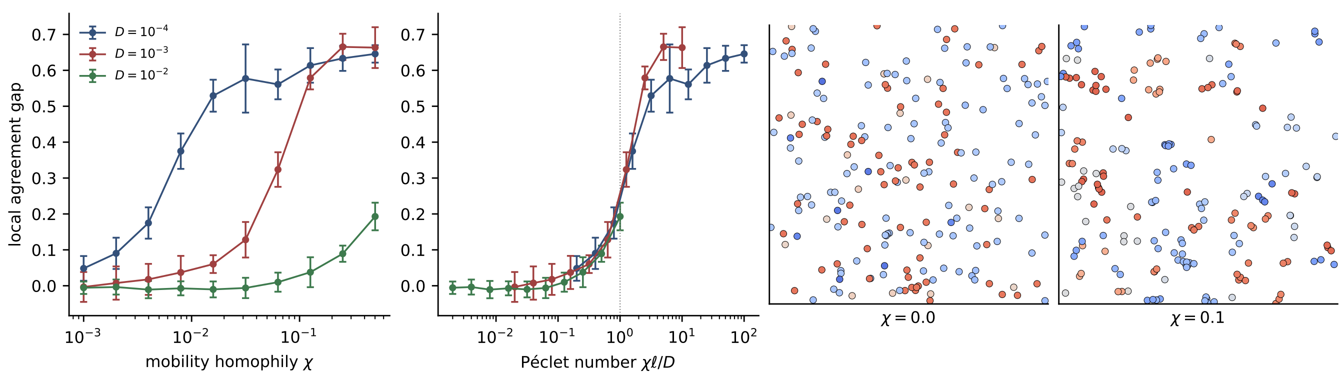

FIG. 11. Level 5: homophilic mobility restores geographic opinion structure. Left: local agreement gap versus the homophilic drift strength χ for three mobilities. Centre-left: the same data against the Péclet number $\chi\ell/D$; the curves approximately collapse, with onset at Pe ~ 1 (dotted). Right: final spatial configurations at $D = 10^{-3}$ without ($\chi = 0$) and with ($\chi = 0.1$) homophilic mobility.

TABLE II. Robustness controls at the reference point ($\lambda = 0.6$, $T = 80$): final polarization $\mathrm{Var}(x)$, mean ± s.d. over 12 realisations. The reference row uses 24 realisations.

Variant	Similarity	Neutral	Controversy
Reference	0.66 ± 0.10	0.98 ± 0.03	0.89 ± 0.03
$s_i = 1$	0.68 ± 0.08	1.00 ± 0.01	0.91 ± 0.02
$\kappa = 4$	0.67 ± 0.09	0.99 ± 0.01	0.91 ± 0.02
Reflecting	0.62 ± 0.12	0.97 ± 0.04	0.89 ± 0.03
$k = 5$	0.54 ± 0.08	0.98 ± 0.02	0.90 ± 0.03
$k = 20$	0.76 ± 0.11	0.98 ± 0.03	0.92 ± 0.03
$\rho = 1$	0.87 ± 0.08	0.98 ± 0.02	0.91 ± 0.03
$\rho = 20$	0.54 ± 0.10	0.97 ± 0.04	0.88 ± 0.05

E. The radicalization crossover and the well-mixed limit

Figure 9 compares the simulated $\mathrm{Var}(x)$–λ curves with the theory of Sec. III. The neutral and controversy platforms jump to their saturated values already at $\lambda = 0.05$, the finest nonzero attention share sampled—consistent with the predicted crossovers $\lambda_c \approx 0.01$–0.02, which lie below the grid resolution, and inconsistent with the naive uniform-state thresholds (0.34, 0.55); their curves are essentially independent of system size (centre panel, shown for the neutral platform), as expected for a finite-horizon crossover rather than a phase transition. The similarity platform grows smoothly along λ toward a partial radicalization, consistent with $t_{\mathrm{rad}} \approx 78/\lambda$ against the horizon $T = 80$; a weak, non-monotone trend with N (right panel), at the edge of statistical resolution, is consistent with a partial averaging-out of slot-level fluctuations—with only $k = 10$ slots, occasional high-influence cross-bloc sources produce bursts of repulsion that the population-wide weighting of Eq. (13) smooths away, which is also why the well-mixed curve (solid blue) sits below the ABM points while tracking their shape. The well-mixed integration reproduces the neutral curve nearly quantitatively at all λ and the controversy curve for $\lambda \gtrsim 0.4$ (at smaller λ it underestimates the controversy platform by up to ≈ 0.28, for the same slot-fluctuation reason), with no parameters fitted to these data.

F. Phase diagram: attention neutralises geography

Figure 10 assembles the global picture in the (D, λ) plane, computed for the similarity-driven design—the slowest to radicalize, so the boundaries below are conservative; the other designs saturate at far lower λ. At $\lambda \approx 0$ the physical layer decides the outcome and the Level-1 crossover is recovered: frozen local fragmentation at low D, the mean-field cluster set at high D. As λ grows the platform captures the attention budget, the repulsive channel activates through the residual cross-cutting exposure, and both polarization and extremism rise across *all* mobilities: the high-λ region is a bimodal, radicalised state essentially independent of D. Once the platform curates most of an agent's exposure, it no longer matters how fast people move: geography sets the outcome only for societies whose attention it still owns.

G. Geography: a null result and its repair (Levels 1 and 5)

The bottom panel of Fig. 2 contains a null result that qualifies a common intuition. Moran's I—and the local agreement gap—never departs from noise level, *even in the frozen regime where interactions are strictly local.* The first-order part of this statement is a symmetry: on the torus, the initial law and the dynamics with $\chi = 0$ are translation invariant, so $\mathbb{E}[x_i \mid \mathbf{r}_i = \mathbf{r}]$ is independent of $\mathbf{r}$ at all times—no deterministic opinion map can exist. Symmetry does not forbid *pair*-level structure (nearby agents interacted more), but the measured pair correlations also stay at noise level: the ensemble mean of Moran's I is compatible with zero at every D (largest excursion 2.2 standard errors among eleven mobilities at 6 seeds each, so the null would not detect a weak true correlation), with an across-seed spread of ≈ 0.13, and the local agreement gap remains at 0.018 ± 0.027 even in the frozen regime—an order of magnitude below the 0.4–0.7 signal that develops once opinion–position coupling is switched on (Fig. 11). Disagreeing agents interleave at the same locations, interacting with disjoint compatible subsets, because nothing couples opinion to position. Locality controls *how many* opinions survive but not *where* they live. Claims that low mobility alone produces geographic echo chambers therefore require an opinion–position coupling, however weak.

Level 5 quantifies how much coupling is needed. With the homophilic drift of Eq. (1) ($\chi > 0$: approach compatible neighbours, avoid incompatible ones [23, 24]), spatial opinion domains appear (Fig. 11). The control parameter is the Péclet number comparing drift and diffusion over the interaction range,

$$\mathrm{Pe} = \frac{\chi \, \ell}{D}, \tag{14}$$

and the local agreement gap for different mobilities approximately collapses onto a single curve in Pe, rising from near noise level for $\mathrm{Pe} \lesssim 0.3$ to strong spatial segregation for $\mathrm{Pe} \gtrsim 3$. Geographic opinion structure is thus a threshold phenomenon in the strength of opinion-dependent mobility—and the Brownian model of Levels 1–4 sits below the threshold by construction, making it the correct null baseline.

V. DISCUSSION

a. Summary. We combined bounded-confidence opinion dynamics, Brownian and homophilic mobility, and an algorithmically curated digital layer under a finite attention budget, and found that the interaction between platform design and influence psychology—not either alone—decides the collective outcome. With assimilative influence, the neutral platform is a defragmenting force and homophilic curation builds echo chambers; with a repulsive channel, every design radicalizes, the neutral platform sets the ceiling, and homophilic curation is a buffer. The two-bloc reduction, the finite-horizon crossover $\lambda_c(T)$, and the well-mixed limit make these statements analytical rather than merely observational, and the robustness and finite-size scans show they are bulk properties of the model. A cautionary methodological note falls out for free: the naïve linear-stability analysis of the uniform state predicts thresholds an order of magnitude too large, because fragmentation preempts it—in coupled opinion–network systems, the state that matters for stability is the one the fast dynamics builds first.

b. Relation to experiments. The inversion offers a candidate population-scale mechanism consistent with the field experiment of Bail *et al.* [4], in which a month of bot-curated exposure to opposing views made Republican participants substantially more conservative (the corresponding shift for Democrats was not statistically significant—an asymmetry that our symmetric, one-dimensional model does not address): in our terms, an intervention that raises cross-cutting exposure moves a similarity-curated population toward the neutral ceiling, and when the repulsive channel dominates, that movement increases polarization. We stress that this comparison is qualitative—no model parameter is calibrated to data—and that the repulsive channel itself is empirically contested (Sec. II): the supporting longitudinal evidence [9] concerns a single non-political preference item (adolescents' taste in clothing style), finds little support for bounded confidence itself, and implies only low macro-level polarization in its own calibrated simulations, while controlled experiments find little evidence of negative shifts at all [31]. The field-experimental record on exposure interventions is equally mixed: counter-attitudinal exposure has also *reduced* affective polarization [5], and large platform experiments altering feed composition found little attitudinal effect [6]. Read this way, the model turns a psychological question into a measurable platform-policy question: it predicts intervention effects of *either* sign depending on the prevalence of negative influence in the target population—consistent with, though certainly not established by, the heterogeneous field record. When influence is assimilative, added exposure diversity heals fragmentation (Sec. IV A); interventions on recommendation algorithms that ignore which influence regime the population is in can therefore backfire in either direction. At the collective level, our neutral-platform result reaches, by a different microscopic route—negative influence on a single opinion dimension rather than partisan sorting across many identity dimensions—the same conclusion as Törnberg [32]: digital media polarize not by isolating users in echo chambers but by exposing them to interaction beyond their local bubble.

c. The geography null result. Under opinion-independent mobility we detect no geographic opinion structure to destroy, within the resolution of our simulations: position and opinion stay uncorrelated at every mobility, by symmetry at first order and at pair level within the stated statistical power. Within this model, robust geographic opinion structure therefore requires an opinion–position coupling—homophilic migration, socially structured mobility, or spatially correlated exogenous influence—and Level 5 shows the coupling need only exceed a modest Péclet threshold $\chi\ell/D \sim 1$. This sharpens, rather than contradicts, the mobile-agent literature [20, 23]: what matters is not whether agents move, but whether their movement knows about opinions.

d. Limitations and outlook. Opinions are one-dimensional and confidence bounds static; adaptive-confidence extensions are natural [30]. Opinions also live on a bounded interval, so "radicalization" here means pinning at the boundary of opinion space; in unbounded formulations [3] the analogue is unbounded divergence. The inversion mechanism—the cross-bloc exposure rate $p(y)$—does not depend on this choice, but the saturation values $\mathrm{Var}(x) \approx 1$ do. The two-bloc reduction is heuristic—it assumes symmetric blocs and fast rewiring—and its quantitative success invites a rigorous analysis of the well-mixed dynamics (13)—its McKean–Vlasov limit for $\kappa > 3$, the non-self-averaging regime at heavier tails, and the metastability structure behind the finite-horizon crossover [26, 27]. The attention budget is a fixed split rather than a dynamic allocation; letting λ itself be optimised by an engagement-maximising platform closes an obvious loop. Stubborn agents are implemented but unexplored; sweeping their number and placement against platform designs connects to opinion control [29]. The model also equates the engagement kernel with what agents attend to *and are influenced by*; real engagement optimisation responds to behaviour rather than opinion distance, and exposure need not translate into influence. And the quantitative platform ordering holds at the explored parameters and horizons: at long enough horizons every design with $p > 0$ reaches the same end state—the kinetic point of Sec. III B cuts both ways. Finally, the model's sharpest empirical claim—that exposure diversity helps or harms depending on the prevalence of negative influence—is testable in principle by conditioning field-experiment outcomes on measured latitudes of acceptance.

CODE AND DATA AVAILABILITY

The full simulation code, the scripts that generate every figure and table, the raw sweep data, and animations of the coupled dynamics are openly available at https://github.com/REsteche/social-modeling.

[1] A. Sîrbu, D. Pedreschi, F. Giannotti, and J. Kertész, Algorithmic bias amplifies opinion fragmentation and polarization: A bounded confidence model, PLOS ONE **14**, e0213246 (2019).

[2] F. P. Santos, Y. Lelkes, and S. A. Levin, Link recommendation algorithms and dynamics of polarization in online social networks, Proceedings of the National Academy of Sciences **118**, e2102141118 (2021).

[3] F. Baumann, P. Lorenz-Spreen, I. M. Sokolov, and M. Starnini, Modeling echo chambers and polarization dynamics in social networks, Physical Review Letters **124**, 048301 (2020).

[4] C. A. Bail, L. P. Argyle, T. W. Brown, J. P. Bumpus, H. Chen, M. B. F. Hunzaker, J. Lee, M. Mann, F. Merhout, and A. Volfovsky, Exposure to opposing views on social media can increase political polarization, Proceedings of the National Academy of Sciences **115**, 9216 (2018).

[5] R. Levy, Social media, news consumption, and polarization: Evidence from a field experiment, American Economic Review **111**, 831 (2021).

[6] B. Nyhan, J. Settle, E. Thorson, M. Wojcieszak, P. Barberá, *et al.*, Like-minded sources on Facebook are prevalent but not polarizing, Nature **620**, 137 (2023).

[7] W. Jager and F. Amblard, Uniformity, bipolarization and pluriformity captured as generic stylized behavior with an agent-based simulation model of attitude change, Computational and Mathematical Organization Theory **10**, 295 (2005).

[8] A. Flache, M. Mäs, T. Feliciani, E. Chattoe-Brown, G. Deffuant, S. Huet, and J. Lorenz, Models of social influence: Towards the next frontiers, Journal of Artificial Societies and Social Simulation **20**, 2 (2017).

[9] T. Tang, T. A. B. Snijders, and A. Flache, An empirical and simulation investigation of bounded confidence and negative influence in opinion dynamics using stochastic actor-oriented modelling, Journal of Artificial Societies and Social Simulation **28**, 2 (2025).

[10] M. Starnini, F. Baumann, T. Galla, D. García, G. Íñiguez, M. Karsai, J. Lorenz, and K. Sznajd-Weron, Opinion dynamics: Statistical physics and beyond, arXiv:2507.11521, accepted in Reviews of Modern Physics (2025).

[11] R. Axelrod, J. J. Daymude, and S. Forrest, Preventing extreme polarization of political attitudes, Proceedings of the National Academy of Sciences **118**, e2102139118 (2021).

[12] D. Sabin-Miller and D. M. Abrams, When pull turns to shove: A continuous-time model for opinion dynamics, Physical Review Research **2**, 043001 (2020).

[13] H. F. de Arruda, F. M. Cardoso, G. F. de Arruda, A. R. Hernández, L. d. F. Costa, and Y. Moreno, Modelling how social network algorithms can influence opinion polarization, Information Sciences **588**, 265 (2022).

[14] V. Pansanella, G. Rossetti, and L. Milli, Modeling algorithmic bias: simplicial complexes and evolving network topologies, Applied Network Science **7**, 57 (2022).

[15] F. Cinus, M. Minici, C. Monti, and F. Bonchi, The effect of people recommenders on echo chambers and polarization, in *Proceedings of the Sixteenth International AAAI Conference on Web and Social Media*, Vol. 16 (2022) pp. 90–101.

[16] R. Pal, A. Kumar, and M. S. Santhanam, Depolarization of opinions on social networks through random nudges, Physical Review E **108**, 034307 (2023).

[17] A. Bellina, C. Castellano, P. Pineau, G. Iannelli, and G. De Marzo, Effect of collaborative-filtering-based recommendation algorithms on opinion polarization, Physical Review E **108**, 054304 (2023).

[18] G. Deffuant, D. Neau, F. Amblard, and G. Weisbuch, Mixing beliefs among interacting agents, Advances in Complex Systems **3**, 87 (2000).

[19] R. Hegselmann and U. Krause, Opinion dynamics and bounded confidence: models, analysis and simulation, Journal of Artificial Societies and Social Simulation **5** (2002).

[20] E. E. Alraddadi, S. M. Allen, G. B. Colombo, and R. M. Whitaker, A novel framework to classify opinion dynamics of mobile agents under the bounded confidence model, Adaptive Behavior **32**, 167 (2024).

[21] C. G. Antonopoulos and Y. Shang, Opinion formation in multiplex networks with general initial distributions, Scientific Reports **8**, 2852 (2018).

[22] E. Kurmyshev, H. A. Juárez, and R. A. González-Silva, Dynamics of bounded confidence opinion in heterogeneous social networks: Concord against partial antagonism, Physica A: Statistical Mechanics and its Applications **390**, 2945 (2011).

[23] F. Pasimeni, R. Wade, and F. Alkemade, Opinion dynamic and social clustering in a 2D space: An agent based experiment, Computational Economics 10.1007/s10614-025-10961-w (2025).

[24] N. Djurdjevac Conrad, N. Q. Vu, and S. Nagel, Co-evolving networks for opinion and social dynamics in agent-based models, arXiv:2407.00145 (2024), arXiv:2407.00145.

[25] J. Lorenz, Continuous opinion dynamics under bounded confidence: a survey, International Journal of Modern Physics C **18**, 1819 (2007).

[26] C. Castellano, S. Fortunato, and V. Loreto, Statistical physics of social dynamics, Reviews of Modern Physics **81**, 591 (2009).

[27] C. Bernardo, C. Altafini, A. Proskurnikov, and F. Vasca, Bounded confidence opinion dynamics: A survey, Automatica **159**, 111302 (2024).

[28] P. Holme and M. E. J. Newman, Nonequilibrium phase transition in the coevolution of networks and opinions, Physical Review E **74**, 056108 (2006).

[29] Y. Tian and L. Wang, Opinion dynamics in social networks with stubborn agents: An issue-based perspective, Automatica **96**, 213 (2018).

[30] G. J. Li, J. Luo, and M. A. Porter, Bounded-confidence models of opinion dynamics with adaptive confidence bounds, SIAM Journal on Applied Dynamical Systems **24**, 10.1137/23M1415503 (2025).

[31] K. Takács, A. Flache, and M. Mäs, Discrepancy and disliking do not induce negative opinion shifts, PLOS ONE **11**, e0157948 (2016).

[32] P. Törnberg, How digital media drive affective polarization through partisan sorting, Proceedings of the National Academy of Sciences **119**, e2207159119 (2022).